TECHFEST · 3D RECONSTRUCTION & ROAMING · ROUND 1

Room3D: Monocular Indoor Reconstruction and a Free-Roaming Web Viewer

Turning a single handheld phone walkthrough into a measurable 3D room (point cloud, floor plan and furniture) that a reviewer can walk through in the browser.

One phone video · CPU only · Metric scale · Floor plan + furniture · First-person roam

1 Project scope

Room3D reconstructs an indoor space from one ordinary phone video. No depth sensor, LiDAR, calibration target, or manual measurement is required: a person simply walks the room once with the camera, and the system recovers the geometry. Every stage runs on CPU only, so it is portable with no GPU dependency. From a single clip it produces a dense colour **point cloud** in real-world metres, the **floor** and **ceiling** planes and vertical **walls** as a clean axis-aligned floor plan, and **furniture** as labelled, oriented 3D boxes carrying real dimensions (width × depth × height). The output is a compact, self-contained scene that the viewer renders with per-element layer toggles and a first-person free-roam mode, so the model is not only viewable but **measurable and explorable**.

KEY CONTRIBUTIONS

- **Metric from monocular.** Per-frame mono-depth is bound to the Structure-from-Motion scale, so the result is in genuine metres, not an arbitrary unit.
- **Clean fusion.** A flying-pixel filter removes the dominant source of monocular-depth noise *before* fusion rather than cleaning up afterwards.
- **Readable layout.** Walls are recovered from tall-cell evidence and snapped to a Manhattan grid, turning a noisy cloud into a straight floor plan.
- **Trustworthy objects.** Furniture detections are merged across many views into complete oriented boxes with measured sizes.

2 Standard Operating Procedure (capture)

Reconstruction quality is set mostly at capture time, so we define a repeatable SOP any operator can follow to map their own space. The guiding idea is a **single slow loop** around the room with the floor always in view: this gives the geometry stage the baseline and ground contact it needs while staying inside what a handheld clip can support (Figure 1).

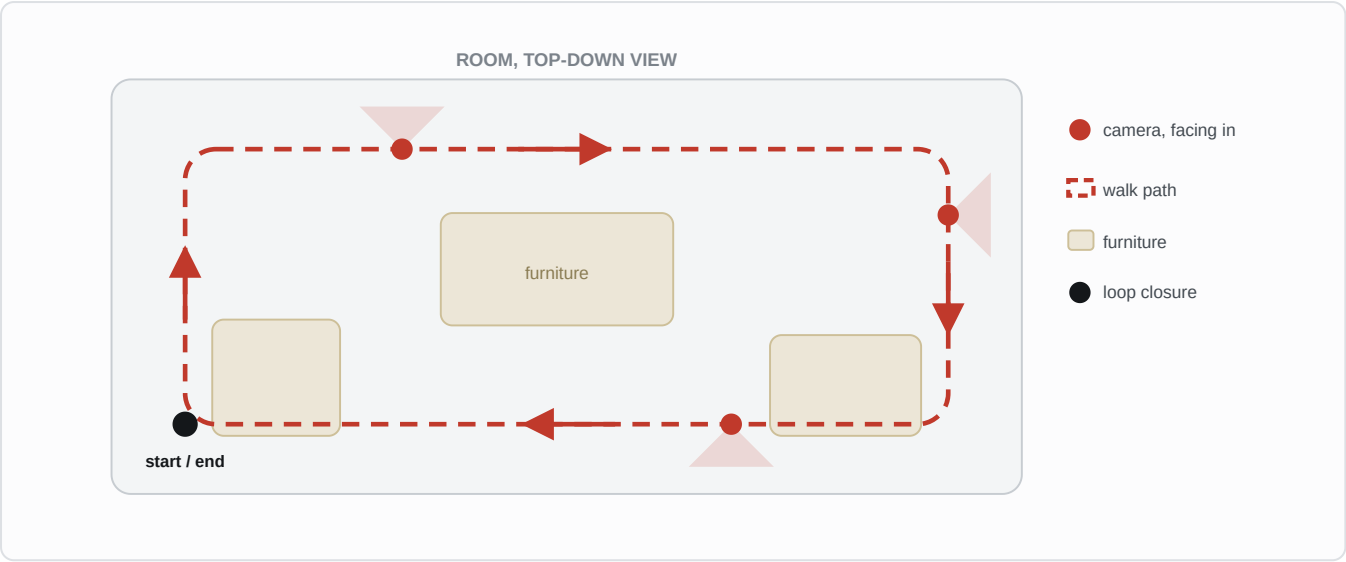


Figure 1. SOP capture pattern. One slow loop along the perimeter, camera facing inward and tilted slightly down, keeps every wall, corner, and the floor in view and closes the loop for a consistent floor plan.

Capture parameter	Recommended setting
Camera	Any phone rear camera, held landscape; let auto-exposure and focus settle before starting.
Resolution / frame rate	`1080p` at `30 fps` (higher is fine; the pipeline down-samples internally).
Trajectory	One closed loop around the perimeter, lens facing the room centre.
Pitch	Tilt slightly downward so a strip of floor is in every frame.
Speed	Slow, steady walk (`~0.3–0.5 m/s`); no abrupt turns or whip-pans.
Clip length	`20–60 s` for a typical room.
Avoid	Motion blur, blank textureless walls, mirrors and glass, and people or pets moving in frame.

3 System architecture

The system has three parts (Figure 2): an offline **reconstruction pipeline** that turns the video into geometry, a compact **scene representation** that stores the result, and a browser **viewer** that renders and roams it. The scene file is deliberately small and explicit so it can be inspected, re-rendered, and opened in standard tools.

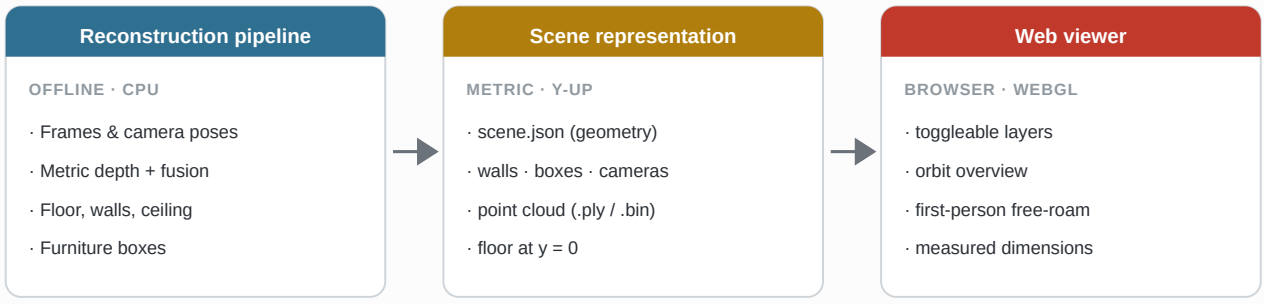


Figure 2. System architecture. The pipeline produces geometry, the scene file stores it in metres with the floor at `y = 0`, and the viewer renders and roams it. The `.ply` cloud also opens directly in MeshLab or CloudCompare.

4 Reconstruction pipeline

The pipeline is a linear sequence of seven stages, each consuming the previous stage's output (Figure 3). Every stage is CPU-friendly and independently inspectable.

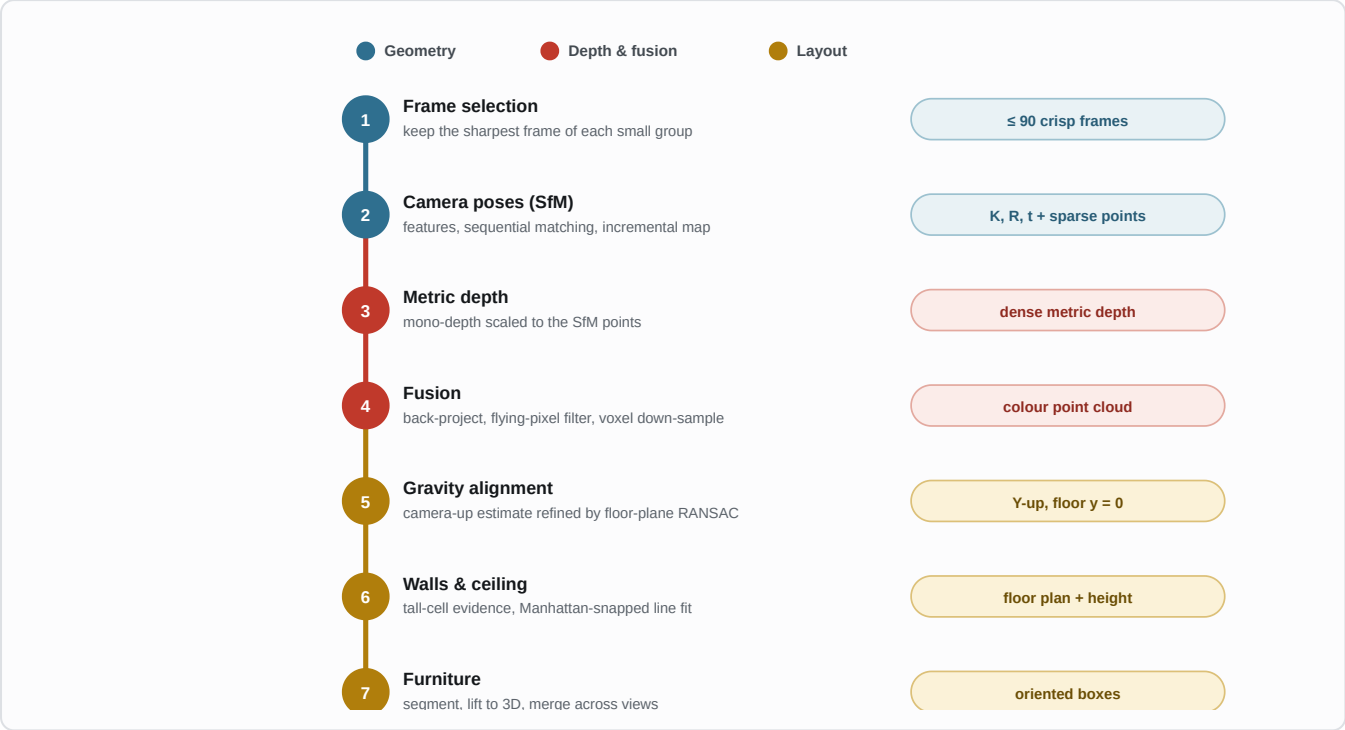


Figure 3. The seven-stage pipeline as a timeline. Colour marks the three phases: geometry (blue), depth and fusion (red), and layout (amber). Each stage's output (right) is the next stage's input.

4.1 Frame selection

Frames are sampled at an adaptive rate so a clip of any length yields up to 90 frames. Within each small group of candidates the **sharpest** frame is kept, measured by the variance of its Laplacian, which discards motion-blurred frames that would poison matching. Frames are resized to a 900 px long side to keep later stages fast.

4.2 Camera poses (Structure-from-Motion)

COLMAP extracts SIFT features under a single shared radial-distortion camera model, matches them *sequentially* across neighbouring frames (the natural ordering of a walkthrough), and runs incremental mapping. Initialisation is deliberately relaxed (handheld walkthroughs have small baselines that defeat the default thresholds), so short, low-parallax clips still register. The stage yields each frame's intrinsics and pose plus a sparse 3D point cloud with 2D-to-3D correspondences used to anchor scale.

4.3 Metric depth

A monocular network (Depth-Anything-V2, indoor-metric variant) predicts a dense depth map per frame. Mono-depth is only correct up to an unknown per-frame scale, so each map is aligned to the SfM points by the **median ratio** between predicted and SfM depth at the reprojected keypoints.

> **KEY IDEA · METRIC SCALE**
> A single **global scale** is shared across all frames, instead of a separate scale per frame. This is the step that makes the reconstruction genuinely metric and keeps depth and poses mutually consistent rather than drifting frame to frame.

4.4 Fusion

Every depth map is back-projected into world space with its intrinsics and pose. A **flying-pixel filter** rejects pixels on steep depth discontinuities (high relative gradient). These are the stray points that monocular depth

smears along the viewing ray and the dominant source of noise. Points beyond 8 m are also dropped, and the combined cloud is voxel-down-sampled at 2.5 cm to a uniform density.

4.5 Gravity alignment

The up direction is estimated from the cameras' own up-vectors, then refined by RANSAC-fitting the dominant horizontal plane beneath the cameras (the floor). The cloud is rotated to a true **Y-up** frame and translated so the floor lies at `y = 0`, making heights, the floor plan, and furniture placement directly meaningful.

4.6 Walls and ceiling

The ceiling height is read from the upper peak of the height histogram. For walls, the floor plan is binned into cells and any cell whose points span a **tall vertical range** is taken as wall evidence. The room's dominant orientation is estimated from the gradients of that floor-plan image, and wall segments are fitted as RANSAC lines **snapped to the Manhattan directions** (exploiting the fact that indoor rooms are mostly axis-aligned), turning noisy evidence into a clean, straight floor plan.

4.7 Furniture

A prompt-free open-vocabulary segmenter (YOLOE) masks furniture in every frame; masks are cleaned by erosion and a depth-median gate that removes background bleed, then back-projected to 3D and lifted into the aligned frame. Observations of the same object across frames are **merged** by label and proximity, and each object is fitted with an oriented bounding box (minimum-area rectangle in plan, snapped to the room yaw). Synonymous detector labels collapse to a canonical vocabulary, so one real object becomes exactly one box with its measured footprint, height, and name.

5 Technical approach & design decisions

- **Monocular over multi-view stereo.** Dense MVS is slow on CPU and fails on the textureless walls that dominate indoor scenes; a learned metric-depth prior fills those regions densely.
- **Global scale sharing.** Per-frame scale drifts; one median scale keeps the floor flat and the cloud coherent across the whole walk.
- **Flying-pixel rejection.** The largest noise source in monocular fusion is edge streaking; a relative-gradient mask removes it cheaply, before fusion.
- **Manhattan floor plan.** Snapping segments to one dominant yaw converts scattered evidence into straight, right-angled walls that read as a real plan.
- **Multi-view box fusion.** Any single mask is partial; merging detections across frames yields complete, stable boxes with trustworthy sizes.
- **CPU-only throughout.** Every stage runs without a GPU, making the system portable and reproducible on commodity hardware.

6 Free-roaming web viewer

The viewer renders the scene in the browser with Three.js / WebGL directly from the compact representation. Each element is a toggleable layer (point cloud, walls, floor, ceiling, furniture boxes, labels, and the camera path), so a reviewer can isolate exactly what they want to judge, and furniture boxes display their measured dimensions.

Two navigation modes are provided. **Orbit** gives a whole-room overview, with double-click to recentre and a top-down floor-plan view. **Free-roam** drops the viewer into a first-person walk at eye height (1.6 m): `W A S D` or arrow keys move, mouse-drag looks around, and `Space` / `Ctrl` change height. Movement is blocked by the reconstructed walls and kept inside the room, so a reviewer can walk through the reconstructed space as if standing inside it, directly satisfying the roaming requirement.

FIDELITY & LIMITATIONS

Because depth is monocular, accuracy is centimetre-level rather than survey-grade, and reconstruction degrades on reflective or glass surfaces, very dark rooms, and large blank walls with no texture for the pose stage. The SOP in Section 2 is designed to steer capture away from exactly these failure modes, which is why capture discipline is treated as a first-class part of the method.

Built on COLMAP (poses), Depth-Anything-V2 (metric depth), YOLOE (segmentation), and Three.js (viewer). Model geometry is stored in metres, Y-up, with the floor at `y = 0`.